

Meiru Zhang

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Research areas: Large Language Model, Robustness and Interpretability, Agent, Event-Centric AI, Neurosymbolic AI

EDUCATION

- University of Cambridge**, PhD in Language Technology Lab Oct 2021 – Aug 2026
- *Supervisor:* Prof. Nigel Collier | **Gates Cambridge Scholar** (~3% acceptance) | *Research:* Event Extraction/Forecasting, RAG, LLM Agents
 - *Thesis:* **Event-Centric AI: From Oracle-Free Event Extraction to Efficient Forecasting**
- Imperial College London**, MSc in Computing (Machine Learning) Sept 2018 – Sept 2019
- **Distinction** | *Research:* Personalization, Active Learning, Reinforcement Learning
 - *Thesis:* **Multimodal Active Learning from Human Data** — learn-to-learn for data-scarce engagement prediction in autism clinical sessions.
- University College London**, BEng in Electronic and Electrical Engineering Sept 2015 – Jun 2018
- **GPA:** 81 (Top 5%) | **Dean's List** | Goldsmith Sessional Prize (2016, 2017)

WORK EXPERIENCE

- Epic Games, London, UK**, ML Research Intern (Part-time) Feb 2026 – Present
- **Task:** Improve the AI Assistant for Unreal Engine and its Code Generation Agent, where LLM agents struggle with tool use in domain-specific, low-resource programming environments.
 - **Action:** Finetuning and post-training code generation models; developing evaluation harnesses and feedback mechanisms for the code generation agent; improving intent understanding and output accuracy for game development workflows.
- Huawei Noah's Ark Lab, London, UK**, Research Intern in NLP Feb 2025 – Sept 2025
- **Task:** Improve retrieval-augmented theorem autoformalization with LLMs, addressing under-specified retrieval queries and weak guidance on how to use retrieved premises.
 - **Action:** Built a retrieval-augmented autoformalization pipeline (DRIFT): decompose queries, retrieve premises and illustrative examples with a finetuned BGE-M3 retriever.
 - **Result:** Up to 13.3 F1 improvement over baselines on retrieval; 42.3% absolute improvement on downstream autoformalization (BEq+@10). **Paper accepted to ICLR 2026.**
- Antobot**, Computer Vision Engineer Oct 2018 – Sept 2021
- **Task:** Deploy robust fruit detection, tracking, and yield estimation on edge devices for autonomous field robotics.
 - **Action:** Delivered end-to-end perception stack: curated instance-segmentation dataset, trained real-time detection/tracking models, and built regression modules for yield and fruit size estimation.
 - **Result:** Production-deployed on-device perception pipeline using TensorRT and ONNX on edge hardware.

SELECTED PUBLICATIONS

- **Failure Modes in Multi-Hop QA: The Weakest Link Law** ACL 2026
Meiru Zhang, Zaiqiao Meng, Nigel Collier.
- **DRIFT: Decompose, Retrieve, Illustrate, then Formalize Theorems** ICLR 2026
Meiru Zhang, Philipp Borchert, Milan Gritta, and Gerasimos Lampouras.
- **The Power of Simplicity in LLM-Based Event Forecasting** ACL 2025 Workshop
Meiru Zhang, Auss Abboud, Zaiqiao Meng, and Nigel Collier.
- **Can We Instruct LLMs to Compensate for Position Bias?** EMNLP 2024
Meiru Zhang, Zaiqiao Meng, and Nigel Collier.
- **BAND: Biomedical Alert News Dataset** AAAI 2024
Zihao Fu, Meiru Zhang, Zaiqiao Meng, Yannan Shen, David Buckeridge, Nigel Collier.

- **COFFEE: A Contrastive Oracle-Free Framework for Event Extraction** ACL 2023 Workshop
Meiru Zhang, Yixuan Su, Zaiqiao Meng, Zihao Fu, and Nigel Collier.
- **Multi-modal Active Learning From Human Data: A Deep RL Approach** ICMI 2019
Ognjen Rudovic, Meiru Zhang, Bjorn Schuller, and Rosalind Picard.

SELECTED RESEARCH

Failure Modes in Multi-Hop QA: The Weakest Link Law Aug 2025 – Jan 2026

- **Problem:** Multi-hop QA degrades significantly in long contexts due to position bias and retrieval bottlenecks.
- **Action:** Proposed *Multi-Focus Attention Instruction* (MFAI); evaluated 5 LLMs on MuSiQue and NeoQA.
- **Result:** Identified recognition failure as the core bottleneck; MFAI improves accuracy by up to 11.5%.

LLM-Based Event Forecasting with Structured RAG Jul 2024 – Mar 2025

- **Problem:** LLM-based event forecasting is expensive due to iterative agentic reasoning (e.g., ReAct) and noisy context.
- **Action:** Replaced multi-step agents with single-step RAG; designed compact, structured retrieval formats.
- **Result:** Structured evidence beats raw headlines at ~10% of ReAct inference cost.

Instructing LLMs to Compensate for Position Bias Dec 2023 – May 2024

- **Problem:** In-context learning is unstable under position bias in LLMs.
- **Action:** Proposed **Attention Instruction** to steer attention; evaluated 6 instruction designs across GPT-4, LLaMA-2, and Mistral.
- **Result:** Demonstrated that LLMs can follow regional attention control through instruction design, improving in-context learning stability.

Oracle-Free Event Extraction (COFFEE) May 2022 – Jan 2023

- **Problem:** Event extraction pipelines rely on oracle trigger annotations that do not generalize.
- **Action:** Introduced the Oracle-Free Event Extraction (OFEE) setting; developed the COFFEE framework using contrastive learning.
- **Result:** Enabled event extraction without oracle triggers; established a more realistic evaluation protocol adopted by follow-up work.

Multi-modal Active Learning from Human Data Mar 2019 – Aug 2019

- **Problem:** Labeling multimodal human interaction data is costly; models need to learn efficiently from limited annotations.
- **Action:** Built multimodal user-state models and implemented a DQN-based active learning agent to select the most informative samples for labeling.
- **Result:** Reduced annotation cost by actively prioritizing high-information training examples. Published at ICMI 2019 (with MIT Media Lab).

INDEPENDENT PROJECT

Priaxis — Autonomous Personalised Agent System 2025 – Present

- **Challenge:** How can an LLM agent autonomously manage personal goals while adapting to individual user context without explicit instruction?
- **Approach:** Building a multi-agent system using Claude Code SDK and MCP protocol for structured tool use, with personalised planning and execution grounded in user history and preferences.

SKILLS

Programming: Python, Java, C

ML/NLP: PyTorch, Transformers, vLLM, LangChain, LangGraph, Weights and Biases

Infra: ElasticSearch, TensorRT, OpenCV